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Research statement

My research audits large language models with the measurement methods of psychology, asking whether what a model produces about people holds up against real population data. Most of the work tests this in mental-health assessment, comparing simulated patients and model judgements with survey-weighted NHANES anchors and scoring bias on continuous severity scales. Alongside the audits I fine-tune and evaluate small local models, compare retrieval with fine-tuning on private documents, and use sparse autoencoders to test interpretability claims. Every study ships with a public repository. Both arXiv preprints carry published changelogs withdrawing earlier findings that did not survive my own audits of the ground truth behind them.

Preprints

Plausible Patients, Impossible Populations: Auditing Epidemiological Fidelity in Large Language Model Mental Health Simulations

arXiv:2604.17359, 2026. v2 live 5 August 2026 with re-derived NHANES ground truth and a changelog withdrawing four v1 findings. github.com/pskeough/plausible-patients

28,800 simulated clinical assessments across four models and 120 demographic cohorts, benchmarked against survey-weighted PHQ-8 anchors re-derived from NHANES microdata. Every benchmarkable group comes back inflated by 2.8 to 5.5 points.

The Granularity Gap: A Multi-Dimensional Cross-Generational Audit of Sycophancy in Gemini Models

arXiv:2606.05183, 2026. v3 live 28 August 2026; v2 withdrew three v1 claims. github.com/pskeough/The-Granularity-Gap

8,830 responses from eight variants across three Gemini generations, scored by a judge that agrees with the consensus of 236 human annotations at Cohen's $\kappa = 0.78$ (0.58 between the human raters). The judge's own refuse-or-comply verdict explains 29% of the variance in its continuous severity scores, and no function of that verdict explains more than 35%.

Working papers

A Validity Ladder for LLM-Simulated Clinical Populations

In preparation for *Behavior Research Methods*. github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations

A four-level protocol for checking a simulated population against a real one, from the coherence of a single respondent to the structure of the whole population, validated with planted failures and a pseudo-model built from NHANES respondents, which passes levels 1 and 3. On the Plausible Patients corpus every model sits 2.1 to 4.7 PHQ-8 points above NHANES after post-stratification, and income gradients come back steeper than NHANES in every model (2.1 to 4.8 times pooled).

Private Documents, Local Models: Retrieval, Fine-Tuning and Judging

With Professor Robert H. Tai, Australian Catholic University. In preparation.

Three decisions facing an organisation that answers questions over a private document collection on one consumer GPU: whether a local model can replace a hosted one, whether to use retrieval or fine-tuning, and whether a local model can judge the result. Given the same retrieved passages, GPT-5.1 and the local 8B recover the same share of the gold answer (mean recall 0.712 each), and net judge preference for the hosted models is +0.14 over a local 3.8B and +0.24 over the 8B.

Same Bias, Different Mechanisms: Socioeconomic Cues in LLM Mental-Health Assessment

2026. github.com/pskeough/psych_scope

Counterfactual audit of socioeconomic-cue bias across three model families, paired with a sparse-autoencoder test of whether interpretability accounts for the effect the audit measures.

Telling More Than They Can Know: Verbal Reports on Model Processes in Clinical Scoring

2026. github.com/pskeough/telling-more-than-they-can-know

Explanation faithfulness across four models, with 1,430 scores and 5,594 explanations under a three-vote judge consensus. Socioeconomic status drives 48.75% of score variance and receives 17.2% of the models' own feature-attribution weight.

Deeper Than the Guardrails: Demographic Bias Survives Refusal-Direction Removal in Standardized Psychiatric Screening

2026. github.com/pskeough/deeper-than-the-guardrails

Ten base and ablated model pairs, 95,920 observations. Over-pathologization falls 2.13 PHQ-8 points under ablation in 21 of 21 comparisons and the benchmarkable cisgender cells still sit about 8 points above the NHANES Low-SES mean.

Per-Author LoRA Adapters: Style, Brief Adherence and Corpus Size Across Two Model Families

Manuscript in preparation, 2026.
13 authors, Qwen3 and Ministral 3 at four parameter counts, 110 training runs and 40,726 generated passages against 546 held-out briefs. The adapter moves output toward the author’s style and away from the brief at every model size, the adherence cost shrinks once the training corpus passes roughly 50 to 110 briefs, and a 4B adapter matches three hosted frontier models on the style measures above about a hundred briefs.
Also in preparation with Professor Tai: a two-axis reliability method for LLM-assisted qualitative coding, and a reliability-validity dissociation on real interview data.

Methods and tools

- **Evaluation design.** LLM-as-judge pipelines with continuous multi-axis rubrics, cross-model and cross-lab judge validation against human annotation, blind and order-controlled presentation, planted-failure and positive controls.
- **Statistics.** Mixed-effects models, non-parametric and exact tests, cluster bootstrap, BH-FDR correction, power analysis, survey-weighted estimates from primary microdata, inter-rater reliability. Estimators are implemented from their definitions and checked against reference implementations before use (Krippendorff’s α to under 1e-10 across 200 random matrices, Fleiss’ κ against statsmodels, ICC against Shrout and Fleiss 1979).
- **Models and engineering.** Python for all analysis pipelines; local fine-tuning (SFT, DPO, LoRA) and adapter serving; sparse-autoencoder interpretability; ablation; hybrid dense and lexical retrieval; cross-family replication on rented GPUs with hash-verified provenance.
- **Software.** *Mimesis* (github.com/pskeough/Mimesis_Voice_Clone), an MCP server that drafts in a single author’s style and selects among candidates by distance to a calibrated stylometric fingerprint. *Claude Mind*, a local hybrid-retrieval memory system for agents (TypeScript, SQLite vector search, cross-encoder reranking).

Education

B.A. Psychology, University of Washington, Seattle June 2025
GPA 3.77/4.00, 180 credits, Dean’s List. Undergraduate Teaching Assistant, PSYCH 315, Understanding Statistics in Psychology.

Service and leadership

Students for Sensible Drug Policy, Executive Policy Board (national) 2022–2023
One of two national student representatives, voting on state and federal harm-reduction policy proposals for the national membership.

Students for Sensible Drug Policy, University of Washington chapter lead 2022–2024
Founded UW’s first naloxone distribution station and ran overdose-response trainings with the Washington State Department of Health.

Writing and athletics

Transubstantiation, short fiction with a process note 2026
Submitted to *Ensemble Park*, issue 3.

The Basilisk, novel in revision 2026
Literary post-apocalyptic novel, current draft about 71,000 words.

The Second Tree, essay collection 2026
31 philosophical travel essays.

USA Swimming, U.S. Olympic Trials
B-Final, 100m Butterfly.